

DZDRuntimeMapEditor

User Guide, Setup Tutorial, and Script Reference

Offline documentation for Unity Asset Store

Namespace DZDMapEditor | Assembly DZD.MapEditor | Format version 1

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1. Introduction

DZDRuntimeMapEditor is a first-person, fly-camera map editor that runs in Play Mode. You place props, sculpt voxel terrain, save local .tabsmmap files, undo edits, and bake a finished map back into the open Unity scene. This documentation is the offline user guide and script reference for the asset.

The product lives under Assets/DZDRuntimeMapEditor/. Runtime scripts use the DZDMapEditor namespace and the DZD.MapEditor assembly. Editor menus appear under Tools, Assets, and GameObject as DZDRuntimeMapEditor. Create config assets from Create > DZDRuntimeMapEditor.

Odin Inspector is optional. If Odin is present, Inspector fields use grouped Chinese labels. If Odin is not present, compatibility stubs keep the project compiling. After you remove Odin, clear leftover ODIN_INSPECTOR scripting defines from Player Settings.

2. Requirements

Install and enable these Unity packages before you use the asset:

- Universal Render Pipeline (com.unity.render-pipelines.universal)
- Input System (com.unity.inputsystem)
- uGUI and TextMesh Pro (com.unity.ugui)

Ghost previews and hover outlines use a bundled URP Overlay renderer feature. Run Tools > DZDRuntimeMapEditor > Setup Overlay Ghost And Outline once so the renderer, rendering layers, and MapEditorOverlaySettings asset are connected.

3. Package Layout

Item	Description
Scripts/Runtime	Play Mode systems in namespace DZDMapEditor.
Scripts/Editor	Menus, importers, overlay setup, and scene wiring.
Prefabs	Flyer, UI, terrain brush shells, and placeable item prefabs.
Data	ScriptableObject catalogs and configs (placement, terrain, save, UI).
Scenes	StartScene.unity (start menu) and DefaultScene.unity (editor scene).
Documentation	This offline user guide (PDF or RTF).

Right-click Assets/DZDRuntimeMapEditor and choose Export Package when you ship a sample. Do not

include Odin Inspector or unrelated art packs from elsewhere in the project.

4. Setup Guide

Follow the steps in this section in order. The included scenes are already wired. Use the later steps when you add the editor to a new URP scene.

4.1 Import the asset

1. Import DZDRuntimeMapEditor into a Unity 6 URP project.
2. Confirm the Input System package is installed and that Active Input Handling is set to Input System Package or Both.
3. Confirm the project uses URP. The Overlay feature is a Universal Renderer Feature.
4. If you do not use Odin Inspector, leave the compatibility stub scripts in place and remove any leftover `ODIN_INSPECTOR*` defines.

4.2 Open the included sample

1. Open Assets/DZDRuntimeMapEditor/Scenes/StartScene.unity.
2. Press Play.
3. Choose New Map to enter the editor scene with a seeded ground and an empty item list, or Open Map to load a .tabsmmap file.
4. You can also open DefaultScene.unity directly if you want to skip the start menu.

The sandbox catalog under Data/Placement includes forest and mushroom items. Terrain brushes under Data/Terrain include a sphere and a cube.

4.3 Add the editor to your own scene

Use a URP scene that already has lighting, a ground collider, and a camera you are willing to replace with the flyer. Exit Play Mode before you run the menus.

1. Open your scene.
2. Choose GameObject > DZDRuntimeMapEditor > Map Scene Anchor, or Tools > DZDRuntimeMapEditor > Create Map Scene Anchor. This creates the item root, terrain root, and catalog references.
3. Select the Map Scene Anchor. Assign a Placeable Catalog, Terrain Sculpt Config, and Map

Persistence Config if the menu did not fill them.

4. Choose Tools > DZDRuntimeMapEditor > Add FirstPersonFlyer. This adds the fly camera, placement, sculpt, pause, warehouse, and save wiring.
5. On TerrainSculptController, assign the scene ground as the seed ground if you want New Map to fill the work volume from that collider.
6. Save the scene and press Play.

4.4 Set up ghost fill and hover outline

1. Choose Tools > DZDRuntimeMapEditor > Setup Overlay Ghost And Outline.
2. Confirm your URP Renderer now includes the Map Editor Overlay feature.
3. Confirm MapEditorOverlaySettings exists (Create > DZDRuntimeMapEditor > Overlay Settings if you need a new asset) and that PlacementConfig and TerrainSculptConfig both reference it.

Hover outline and ghost fill use rendering layers HoverOutline and GhostFill. The Overlay feature draws a screen-space silhouette fill and a constant-pixel-width outline. You do not need a third-party outline asset.

4.5 Convert models to placeable items

1. Put prefabs under a folder such as Assets/DZDRuntimeMapEditor/Prefabs/Items/Trees. The folder name can match PlaceableCategoryDef.prefabFolderName so Convert to Placeable assigns the right backpack category.
2. Select the prefabs or the folder.
3. Choose Assets > DZDRuntimeMapEditor > Convert to Placeable. This creates PlaceableItemDef assets, assigns a category when the folder name matches, and registers the items in the catalog.
4. Optionally choose Assets > DZDRuntimeMapEditor > Bake Placeable Icons, or Bake Placeable Icons (Force Overwrite), to render warehouse icons from the meshes.
5. Keep each item Id stable. Map files look up prefabs by that Id. Changing an Id makes old saves skip those objects.

You can also create definitions by hand: Create > DZDRuntimeMapEditor > Placeable Category, Placeable Item, and Placeable Catalog, then drag items into the catalog list.

4.6 Convert models to terrain brushes

1. Select a mesh prefab that represents the brush shape.

2. Choose Assets > DZDRuntimeMapEditor > Convert to Terrain Brush. This bakes a density volume, creates a TerrainBrushDef, and registers it in the brush catalog.
3. Choose Height Column fill for hills and ground slabs, or Solid Interior for closed meshes.
4. To rebake every brush in the catalog, choose Tools > DZDRuntimeMapEditor > Bake Terrain Brush Volumes.

4.7 Apply a saved map to the open scene

Use this after you have edited a map in Play Mode and want the result in Edit Mode for lighting, navmesh, or export. The scene must contain a Map Scene Anchor. Exit Play Mode first.

1. Open the target scene.
2. Choose Tools > DZDRuntimeMapEditor > Apply Map File to Open Scene.
3. Pick a .tabsmap file. The tool adds, updates, and removes objects under the item root only. Lights, the skybox, and the flyer stay untouched.
4. Terrain density is written to a TerrainVolumeAsset and meshed in Edit Mode. The next Play session loads that volume instead of reseeding from the ground.

5. Play Mode Tutorial

This section is a step-by-step walkthrough of a typical editing session. Keys listed here are the defaults. You can change every binding at runtime under Pause > Settings.

5.1 Start menu

The start scene shows the mouse and does not lock the cursor.

- Open Map lists local saves, newest first. Each row shows the display name, last write time, file size, object count, and terrain cell size. You can rename or delete a file. Unreadable or newer-format files are marked unavailable.
- New Map loads the editor scene, seeds solid terrain from the scene ground, and starts with no placed items.
- Escape cancels rename or a delete confirm first, then closes the list.

5.2 Flying and the mouse

After you enter the editor scene, the camera locks and hides the mouse. A crosshair appears at the screen

center and hides while backpack, pause, or load UI is open.

- WASD moves on the horizontal plane relative to the view.
- Space rises. Left or Right Ctrl descends.
- Mouse look yaws the body and pitches the look pivot, clamped to about plus or minus 89 degrees.
- Flight has no world collision and no gravity.
- UI that needs the pointer (backpack, pause, save list) holds the mouse. Click empty game view to lock again when no UI is holding it.
- While a ghost item is held, Height Adjust (V) or Rotate Adjust (Caps Lock) can steal the matching mouse axes from look.

5.3 Item placement

Press 1 (or Numpad 1) to enter item placement. The bottom hotbar shows the warehouse.

- Scroll the mouse wheel to highlight an item. F confirms the highlight and starts a ghost preview of a new instance.
- Press B to open the backpack. Click a category icon to take that item. The backpack closes and ghost preview starts.
- Aim at a placed object to show a hover outline. Left-click picks it up as a ghost so you can move, rotate, or scale it.
- The ghost follows the crosshair ray, sits on the hit point, and lifts slightly along the normal. Collision on the preview is disabled so it does not block later rays.
- Left-click places the ghost on a valid surface. Right-click cancels a new preview, or deletes a picked-up object (undoable).
- Q and E yaw around world up. Shift+Q and Shift+E scale uniformly, with min and max multipliers. You can set Q/E to hold-to-repeat or tap-for-a-fixed-step.

Four placement modes can be toggle or hold-to-use. They do not fire while the backpack is open:

- Height Adjust (V): mouse Y raises or lowers the object. The ground hit is frozen. The preview no longer collides with terrain or placed items, so you can hang objects in the air. Height is clamped relative to the drop point.
- Continuous Place (X): after a left-click, the same item stays in hand and keeps rotation, scale, and height offset. Use this to plant a row of trees.
- Snap to Surface (C): the object's up axis follows the hit normal. An item definition can also force this per prefab.

- Rotate Adjust (Caps Lock): mouse X yaws the object and mouse Y pitches it. If Height Adjust is also on, mouse Y is used for height only.

5.4 Terrain sculpting

Press 2 (or Numpad 2) to sculpt. Switching to terrain cancels an item preview and closes the backpack. Voxel density is written inside a work box (default about 128 x 32 x 128 meters, cell size about 0.75 meters) and meshed with collision so you can place items on new slopes.

- New Map fills the seed ground collider as solid, then hides the original ground renderers and colliders.
- If a save already contains terrain, load uses that density and does not reseed.
- Scroll to choose a brush. F resets brush yaw and scale.
- Hold left mouse to add (union). Hold right mouse to subtract. Hold middle mouse to smooth only near the iso surface.
- While you hold a mouse button, stamps follow your own movement, including vertical flight. New terrain does not push the brush up, so you can build cliffs while rising. After you release, the preview snaps back to the ground.
- Q/E rotate the brush. Shift+Q/E scale it.

Mesh options on TerrainSculptConfig include Surface Nets (default), Marching Cubes, and cube faces. Soft-edge skirts, coplanar merge, flat shading, and a separate shadow mesh are also configurable. Dirty chunks remesh with a per-frame budget.

5.5 Save, load, undo, and redo

Map files use the .tabsmmap extension. In the Editor they write to the project Maps/ folder when that option is enabled. In a player build they write to persistentDataPath/Maps/.

- Ctrl+S quick-saves the current file, or Map.tabsmmap if none is open.
- Ctrl+O opens the same load list as the start menu.
- Ctrl+Z undoes. Ctrl+Y redoes. History defaults to 32 steps. Loading a map clears history.
- On macOS, Ctrl shortcuts also accept Command.

Before save or load, the editor cancels a held item preview and ends an unfinished terrain stroke. A save stores placed items (instance Id, definition Id, position, euler, scale), compressed terrain density when present, and optional camera pose.

5.6 Pause menu and settings

Escape opens pause (time scale stops by default). Opening pause cancels previews, ends terrain strokes, closes the backpack, and shows the mouse. Resume, Save Scene, Load Scene, and Settings are on the main pause page.

Escape closes layers from the inside out: key rebind, then Settings, then the save page (canceling rename or confirm first), then the load list, then pause itself.

Settings has four tabs: Control (all key bindings), Character (fly speed and look), Placement (Q/E style, height and rotate feel, scroll invert, mode trigger style), and Terrain (remesh budget, stamp spacing, smooth, brush rotate and scale). Click a key row, press the new key, or press Escape to cancel. Conflicts show the other action name.

6. Default Controls

Item	Description
1 / 2	Item placement / terrain sculpt (numpad included).
WASD, Space, Ctrl	Fly, rise, descend.
Mouse	Look, or height/rotate adjust while those modes are on.
Scroll, F	Warehouse or brush select, then confirm.
B	Backpack.
LMB / RMB / MMB	Place or add / cancel, delete, or subtract / smooth.
Q E, Shift+Q E	Rotate / scale.
V, X, C, Caps Lock	Height, continuous place, snap to surface, rotate adjust.
Ctrl+S / O / Z / Y	Save / load list / undo / redo.
Esc	Pause, then nested UI.

7. Editor Menus

Item	Description
GameObject / Tools > Map Scene Anchor	Create the scene content anchor.
Tools / GameObject > Add FirstPersonFlyer	Add or rewire the fly editor rig.
Tools > Apply Map File to Open Scene	Bake a .tabsmmap into the open scene.
Assets / Tools > Convert to Placeable	Create item definitions from prefabs.
Assets / Tools > Convert to Terrain Brush	Bake a brush volume from a mesh.
Tools > Bake Terrain Brush	Rebake all catalog brushes.

Volumes

Assets / Tools > Bake Placeable Render warehouse icons.

Icons

Tools > Setup Overlay Ghost And Wire URP Overlay, layers, and settings.

Outline

Create extra config assets from Create > DZDRuntimeMapEditor.

8. Save Files

MapPersistenceConfig.CurrentFormatVersion is 1. The on-disk container starts with the bytes TMAP, then a binary version, UTF-8 JSON (MapDocumentJson), then optional GZip-compressed quantized density (0-255). Newer format versions fail to load with an error.

JSON fields include mapName, optional camera pose, item array (instanceId, defId, position, euler, scale), and terrain layout (origin, cells, cellSize, chunkCells). Missing definition Ids are skipped and a toast reports how many objects were dropped.

Saves do not include lights, post-processing, or the skybox. Those stay in the Unity scene. Rename and delete operate on disk files and are not undoable. Load clears undo history.

9. Script Reference

Use this section when you wire scenes in code, extend catalogs, or call save and undo from your own UI. All types below are in namespace DZDMapEditor unless noted. Most gameplay wiring is Inspector data on prefabs and ScriptableObjects, not hardcoded values.

9.1 MapEditorInfo

Static package labels: PackageId, DisplayName, CreateMenu, ToolsMenu, AssetsMenu, and GameObjectMenu. Use these strings if you add your own MenuItem entries next to the built-in ones.

9.2 MapSceneAnchor

MonoBehaviour on the scene. Holds PlacedItemsRoot, TerrainChunksRoot, Catalog, SculptConfig, TerrainVolume, PersistenceConfig, TerrainPresenter, and SculptController. Apply Map File to Open Scene reads this component. BindRuntimeRefs, BindTerrainVolume, BindTerrainPresenter, BindSculptController, and BindContentRoots exist for editor setup scripts.

9.3 Placeable content

PlaceableCatalog is a ScriptableObject list of PlaceableItemDef. TryGet(id, out def) matches PlaceableItemDef.Id or the asset name.

PlaceableItemDef fields: Id (stable save key), DisplayName, Category, Prefab, Icon, AlignToSurfaceNormal, ExtraSurfaceOffset. Create from Create > DZDRuntimeMapEditor > Placeable Item.

PlaceableCategoryDef fields: DisplayName, SortOrder, PrefabFolderName. Importers match Prefabs/Items/<folder> to prefabFolderName.

PlacedItem is the runtime marker on a spawned instance. Definition, InstanceId, Bind, BindInstanceId, EnsureInstanceId, and FindOn(Collider) are the public API. Do not change InstanceId by hand on objects you intend to save.

9.4 Placement runtime

PlacementController runs pick, ghost, warehouse, backpack, and the four placement modes. Assign PlacementConfig, the fly camera, GameplayCursor, OutlineSelectionVisual, item root, hotbar, backpack, PlacementModeState, EditorToolState, and MapEditSession.

PlacementConfig stores ray mask and distance, surface offset, Q/E RepeatKeyStyle, rotate and scale rates, height range, ModeSwitchStyle for V/X/C/CapsLock, overlay settings, and ghost tint. ModeSwitchStyle is Toggle or Hold. RepeatKeyStyle is Hold or Tap.

IPlacementInput.Read(PlacementConfig) returns PlacementInput. KeyboardMousePlacementInput is the default. ISelectionVisual.Show/Hide is implemented by OutlineSelectionVisual.

PlacementMutation describes an undoable place, transform, or delete. MapEditSession.HandlePlacement records PlaceItemCommand, TransformItemCommand, or DeleteItemCommand.

9.5 Terrain sculpt

TerrainSculptController implements ITerrainMapPort. Public methods: TryCapture, ApplySnapshot, ResetToSeeded, BeginPatch, EndPatch, ApplyPatch, CancelStroke. Assign TerrainSculptConfig, TerrainBrushCatalog, EditorToolState, camera, GameplayCursor, chunks root, optional TerrainVolumeAsset, seed ground, hotbar, and MapEditSession.

TerrainSculptConfig holds the work box (WorkCenter, WorkSize), CellSize, ChunkCells, IsoLevel,

Mesher (TerrainMesherType: Cubes, MarchingCubes, SurfaceNets), remesh budget, stamp spacing, smooth iterations and strength, brush rotate and scale keys, and overlay ghost tint.

TerrainBrushCatalog lists TerrainBrushDef. Each def has DisplayName, Prefab, Icon, FillMode (HeightColumn or SolidInterior), baked Resolution, BoundsMin, BoundsSize, IsBaked, Sample, and SetBaked.

TerrainVolumeAsset stores quantized density for Edit Mode and the next Play session.

TryCreateSnapshot and TryWriteSnapshot convert to and from TerrainVolumeSnapshot.

TerrainChunkPresenter remeshes that asset in the Editor.

ITerrainMapPort is the undo boundary: capture or apply a full snapshot, reset to seeded ground, and apply a TerrainDensityPatch. TerrainStampOp is used internally for add, subtract, and smooth.

9.6 Persistence and history

MapPersistenceController: Save, TrySaveAs, TrySaveToPath, LoadFromPath, ShowLoadPanel, HideLoadPanel, TryRenameMap, DeleteMap, NotifyRenamed, NotifyDeleted, ConsumeLoadPanelEscape, IsLoadPanelOpen. Wire MapPersistenceConfig, MapEditSession, optional TerrainSculptController and FirstPersonFlyController, GameplayCursor, PlacementController, backpack, MapLoadPanelView, PauseMenuController, and MapSessionRequest.

MapPersistenceConfig: CurrentFormatVersion, FileExtension (.tabsmap), DefaultFileName, UseProjectMapsFolderInEditor, MaxUndoSteps, SaveCamera, save/load/undo/redo keys, and toast strings.

MapFileStore is the static file API: ResolveDirectory, ResolvePath, SanitizeStem, TryRename, TryDelete, IsMapFile, ListMaps, TryPeek, Write, TryRead, QuantizeDensity, DequantizeDensity.

MapSessionRequest.RequestNew and RequestLoad are how the start menu tells the editor scene what to do. MapPersistenceController consumes the request on Start.

MapEditSession owns MapHistory and PlacedItemWorld. CanUndo, CanRedo, Undo, Redo, ClearHistory, HandlePlacement, and RecordTerrainPatch are the public commands. IPlacedItemWorld covers Register, Unregister, Spawn, Destroy, SetPose, ClearAll, and CaptureAll.

MapDocumentJson and MapItemJson are the serializable save records. MapFileInfo is a list-row view of a file. TerrainVolumeSnapshot and TerrainDensityPatch are memory copies for save and undo.

9.7 Player, tools, overlay, and toast

FirstPersonFlyController uses FirstPersonFlyConfig (horizontal and vertical speed, look sensitivity, invert Y, pitch limits, lock cursor, WASD and rise/descend keys). IFirstPersonFlyInput.Read returns FlyInput. KeyboardMouseFlyInput is the default.

GameplayCursor locks and unlocks the pointer. IsLocked, IsUiHold, RelockedThisFrame, and ForceLock are used by placement, sculpt, and pause so world tools do not fire through UI.

EditorToolConfig stores the 1/2 keys and toast strings. EditorToolController writes EditorToolState.Mode (EditorToolMode.Placement or Terrain). IsPlacement and IsTerrain are helpers.

MapEditorOverlaySettings is the Overlay ScriptableObject: injection point, hover and ghost rendering layers, outline width, resolution scaling, and colors. SetHover and the ghost setters are used at runtime. MapEditorOverlayFeature is the URP renderer feature that draws them.

ToastChannel.Raise(string) or Raise(message, durationSeconds) broadcasts ToastRequest to ToastHudView. Assign the same channel on persistence, tools, and placement configs so the player sees one HUD.

9.8 Typical code calls

Save the current map from your own button:

```
FindFirstObjectByType<MapPersistenceController>().Save();
```

Load a path, then clear history is already done inside LoadFromPath. To apply a density snapshot in Play Mode:

```
sculpt.ApplySnapshot(snapshot);
```

Look up a placeable before you spawn one yourself:

```
if (catalog.TryGet(defId, out var def)) { /* instantiate def.Prefab */ }
```

Prefer Inspector wiring on Map Scene Anchor and the FirstPersonFlyer prefab over FindObjectOfType in shipping code. The snippets above are for editor tools and one-off tests.

10. Known Limits

- Fly mode has no collision with the world and no walking locomotion.
- Terrain edits stay inside TerrainSculptConfig work bounds.
- There is no snap grid or alignment gizmo beyond the crosshair, height adjust, and surface snap.

- Saves do not include lighting, post-processing, or skybox.
- There is no multiplayer editing and no cloud save.
- Ghost fill and hover outline require the Overlay feature after Setup Overlay Ghost And Outline.
- Changing PlaceableItemDef.Id orphans objects in old .tabsmmap files.

11. Support

This file is the offline documentation for DZDRuntimeMapEditor. Keep a public support website and publisher email available as required by the Unity Asset Store Publisher Guide.

For Asset Store reviewers: this package includes scripts that require setup, a numbered table of contents, an English setup tutorial, and a script reference. The mandatory offline format is PDF or RTF.